
Integrative Programming and Technologies III
Test 1 (Inheritance, Abstract classes and Interfaces)

30 March 2023

Duration: 1 Hour 15

Total: 40 marks

Question 1-10: 10 Marks

Indicate whether the following statements are **TRUE** or **FALSE**.

1. In Java, a class can extend **ONLY** one superclass.
2. It is a good programming practice to declare the data members in a superclass in inheritance as public.
3. Method overload occurs between a superclass and subclass in inheritance.
4. Method override occurs in the same class.
5. You can create an object of an abstract class.
6. You can create an object of an interface.
7. In polymorphism, the method which is called depends on the type of object.
8. In polymorphism, an appropriate method of an object is determined when the program is running (i.e., at runtime).
9. A class must be declared abstract if it contains an abstract method.
10. An abstract class can implement an interface.

Question 11 2 Marks

The two main relationships in object-oriented programming are _____ and _____.

Question 12 2 marks

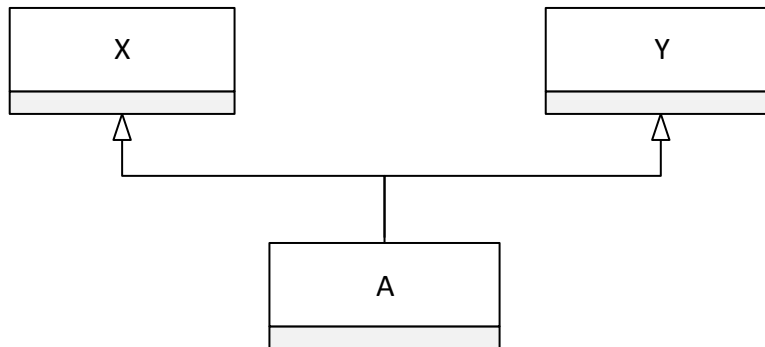
Explain why the **highlighted** code is invalid.

```
public interface IPriceCalculator {  
    public abstract double calcTotalPrice();  
    public abstract int calcTotalPrice();  
}
```

Question 13

3 marks

The diagram below shows the classes **A**, **X** and **Y**. If **X** contains only constants and abstract methods and **Y** contains variables and method implementations, explain (**do not re-draw** the diagram), what can be done to avoid multiple inheritance.



Question 14

5 marks

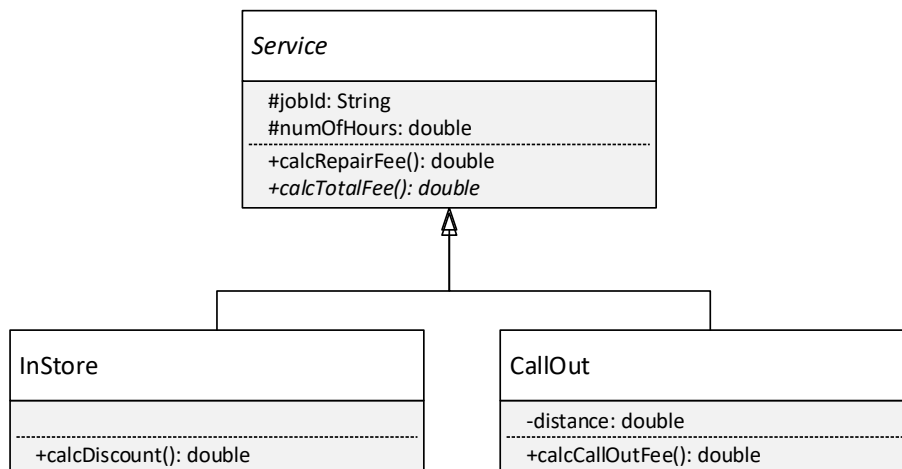
Class **MyClass** implements an interface **IMyInterface**. State whether the following statements are valid or invalid. Explain your answer.

- | | | |
|------|--|-----------|
| i. | <code>MyClass c=new MyClass();</code> | (1 marks) |
| ii. | <code>IMyInterface i1=new IMyInterface();</code> | (2 marks) |
| iii. | <code>IMyInterface i2=new MyClass();</code> | (2 marks) |

Question 15

8 marks

XYZ Locksmith offers two types of services: **in-store** (customers go to the shop with their locks for repair) and **call-out** (the locksmith travels to clients to repair their locks).



The diagram above shows the relationship among the classes **Service**, **InStore** and **CallOut**.

1. Complete the class header of **Service**.

2 marks

```

_____ Service {
}
  
```

2. Write the method declarations for **calcRepairFee()** and **calcTotalFee()** in Service.

2.1 _____	2 marks
2.2 _____	2 marks

3. Complete the class header for class **CallOut**

2 marks

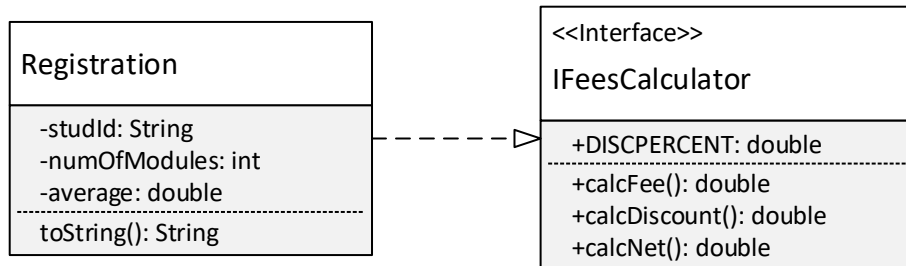
```

_____ CallOut _____ {
}
  
```

Question 16

10 marks

NLM University offers discount on fees to students that obtained an average of 75% or more. The diagram below shows the class **Registration** and interface **IFeesCalculator**. Registration contains a student's registration details: **studId** (student id), **numOfModules** (the number of modules registered) and **average** (the average mark). **IFeesCalculator** contains **DISCPERCENT**: the percentage discount (a constant, 25%); **calcFees()**: returns the fees, based on the number of modules registered (**each module is R12 000**); **calcDiscount()**: returns the discount on fees (based on **DISCPERCENT**); and **calcNet()**: returns the net fees (calculated as the fees less discount on fees).

1. Complete the code for the interface **IFeesCalculator**

5 marks

```

public interface IFeesCalculator {
    _____
    _____
    _____
    _____
}
  
```

2. Complete the class header for **Registration**

2 marks

```

_____Registration _____{
}
  
```

3. Complete the method **calcDiscount()** in **Registration** by calling the **calcFee()**. Assume **calcFee()** has already been implemented.

3 marks

```

public double calcDiscount() {
    _____
    _____
    _____
}
  
```